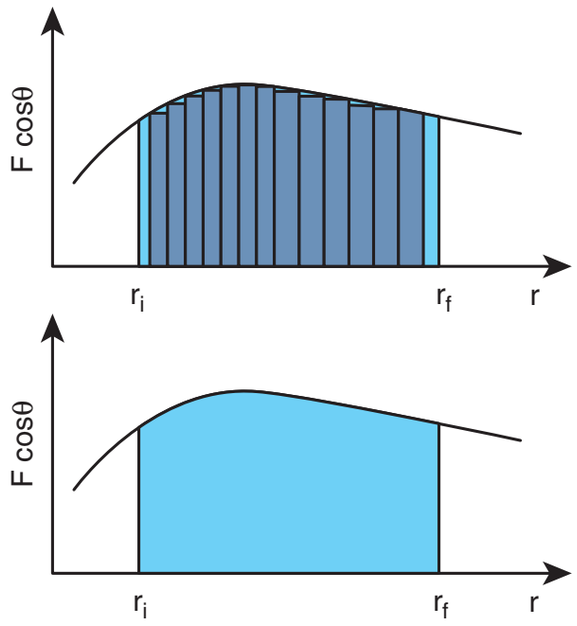


Figure 4.6 Work done by a variable force

EXAMPLE 4.6

A variable force $F = k\,x^2$ acts on a particle which is initially at rest. Calculate the work done by the force during the displacement of the particle from $x = 0$ m to $x = 4$ m. (Assume the constant $k = 1\ \text{N m}^{-2}$)

Solution

Work done,

$$W = \int_{x_i}^{x_f} F(x)\,dx = k\int_{0}^{4} x^2\,dx = \frac{64}{3}\ \text{N m}$$

4.2 ENERGY

Energy is defined as the capacity to do work. In other words, work done is the manifestation of energy. That is why work and energy have the same dimension (ML^2T^{-2})

Work ⇔ Energy

The important aspect of energy is that for an isolated system, the sum of all forms of energy i.e., the total energy remains the same in any process irrespective of whatever internal changes may take place. This means that the energy disappearing in one form reappears in another form. This is known as the law of conservation of energy. In this chapter we shall take up only the mechanical energy for discussion.

In a broader sense, mechanical energy is classified into two types

1. Kinetic energy
2. Potential energy

The energy possessed by a body due to its motion is called kinetic energy. The energy possessed by the body by virtue of its position is called potential energy.

The SI unit of energy is the same as that of work done i.e., N m (or) joule (J). The dimension of energy is also the same as that of work done. It is given by $[ML^2T^{-2}]$. The other units of energy and their SI equivalent values are given in Table 4.2.

Table 4.2 SI equivalent of other units of energy

Unit	Equivalent in joule
1 erg (CGS unit)	10^{-7} J
1 electron volt (eV)	$1.6\text{x}10^{-19}$ J
1 calorie (cal)	4.186 J
1 kilowatt hour (kWh)	$3.6\text{x}10^{6}$ J

UNIT-4(XI-Physics_Vol-1).indd 173 05-01-2022 22:19:36